Module 2

Random variables

Introduction
- random variables
- Probability mass Function, distribution and density functions
- joint Probability distribution and joint density functions
- Marginal, conditional distribution and density functions
- Mathematical expectation, and its properties
- Covariance , moment generating function

Random Experiment: A **random experiment** is a process by which we observe something uncertain. After the experiment, the result of the random experiment is known. An **outcome** is a result of a random experiment.

Example: Throwing a die.

Trial: Any particular performance of a random experiment is called a ***trial***

Sample space: The set of all possible outcomes of a statistical experiment is called the **sample space.** An **event** is a subset of a sample space.

Example: roll a die; sample space: S={1,2,3,4,5,6}. If we would like to know the probability of getting even number, then event is {2,4,6}.

Statistical Definition:

If a random experiment or trail results in ***'n'*** possible outcomes(or cases), out of which ***'m'*** are favourable to the occurrence of an event $\boldsymbol{E}$, then the probability '**p**' of occurrence of E, denoted by $\boldsymbol{P(E)}$, is given by:

$$p = P\ (E) = \frac{Number\ of\ favourable\ cases}{Total\ number\ of\ exhaustive\ cases} = \frac{m}{n}$$

(i) Since $m \geq 0, n \geq 0 \quad and\ m \leq n$

$P(E) \geq 0 \quad and \quad P(E) \leq 1 \qquad \Longrightarrow 0 \leq P\ (E) \leq 1$

(ii) $P(E) + P(\bar{E}) = 1$

Algebra of Events:

For events A, B, C

(i) $A \cup B \;=\; \{\omega \in S: \omega \in A \;\; or \; \omega \in B\}$

(ii) $A \cap B \;=\; \{\omega \in S: \omega \in A \;\; and \; \omega \in B\}$

(iii) $\bar{A} \;=\; \{\omega \in S: \omega \notin A\}$

(iv) $A - B \;=\; \{\omega \in S: \omega \in A \;\; but \; \omega \notin B\}$

(v) $A \subset B \;\; for \; every \; \omega \in A, \;\; \omega \in B$

(vi) $A = B$ if and only if A and B have same elements, $i.e., A \subset B$ and $B \subset A$

(vii) A and B disjoint events (mutually exclusive) $\Longrightarrow A \cap B = \Phi \; (null \; set)$

(viii) $A \cup B$ can be denoted by $A + B$ if A and B are disjoint.

(ix) A Δ B denotes those ω belonging to exactly one of A and B, *i.e.*,

$$A \,\Delta\, B \;=\; \bar{A}\,B \;\cup\; A\,\bar{B} \;=\; \bar{A}\,B + A\,\bar{B} \quad \text{(disjoint events)}$$

Theorems of Probability

1) Addition Theorem: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
2) $P(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \ldots + (-1)^{n-1} P(A_1 \cap A_2 \cap \cdots A_n)$

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

If the events A and B are independent, $P(A|B) = P(A)$

$$P(B|A) = P(B)$$

Multiplication theorem of Probability for Independent events:

$$P(A \cap B) = P(A) \cdot P(B)$$

For n events, $P(A_1 \cap A_2 \cap \cdots \ldots . A_n) = P(A_1)\, P(A_2 | A_1)\, P(A_3 | A_1 \cap A_2) \ldots \ldots \times$

If the events are not independent

$$P(A_n | A_1 \cap A_2 \cap \cdots \ldots . A_{n-1})$$

Example 26.14. *Two cards are drawn in succession from a pack of 52 cards. Find the chance that the first is a king and the second a queen if the first card is (i) replaced, (ii) not replaced.*

Solution. (*i*) The probability of drawing a king $= \frac{4}{52} = \frac{1}{13}$.

If the card is replaced, the pack will again have 52 cards so that the probability of drawing a queen is 1/13.

The two events being independent, the probability of drawing both cards in succession $= \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$.

(*ii*) The probability of drawing a king $= \frac{1}{13}$.

If the card is not replaced, the pack will have 51 cards only so that the chance of drawing a queen is 4/51.

Hence the probability of drawing both cards $= \frac{1}{13} \times \frac{4}{51} = \frac{4}{663}$.

Bayes Theorem of Probability:

Theorem: If $\boldsymbol{E_1, E_2, \ldots\ldots, E_n}$ are mutually disjoint events with $\boldsymbol{P(E_i) \neq 0}$, $(i = 1, 2, \ldots. n)$, then for any arbitrary event $\boldsymbol{A}$ which is a subset of $\bigcup_{\boldsymbol{i=1}}^{\boldsymbol{n}} (\boldsymbol{E_i})$ such that $\boldsymbol{P(A) > 0}$, we have

$$P(E_i|A) = \frac{P(E_i)\, P(A|E_i)}{\sum_{i=1}^{n} P(E_i)\, P(A|E_i)} \;;\; i = 1,2, \ldots\ldots, n$$

Example 26.27. *There are three bags : first containing 1 white, 2 red, 3 green balls ; second 2 white, 3 red, 1 green balls and third 3 white, 1 red, 2 green balls. Two balls are drawn from a bag chosen at random. These are found to be one white and one red. Find the probability that the balls so drawn came from the second bag.*

(J.N.T.U., 2003)

Solution. Let B_1, B_2, B_3 pertain to the first, second, third bags chosen and A : the two balls are white and red.

Now $$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$$

$$P(A/B_1) = P \quad \text{(a white and a red ball are drawn from first bag)}$$

$$= ({}^1C_1 \times {}^2C_1)/{}^6C_2 = \frac{2}{15}$$

Similarly $$P(A/B_2) = ({}^2C_1 \times {}^3C_1)/{}^6C_2 = \frac{2}{5}, \; P(A/B_3) = ({}^3C_1 \times {}^1C_1)/{}^6C_2 = \frac{1}{5}$$

By Baye's theorem, $$P(B_2/A) = \frac{P(B_2)\, P(A/B_2)}{P(B_1)\, P(A/B_1) + P(B_2)\, P(A/B_2) + P(B_3)\, P(A/B_3)}$$

$$= \frac{\frac{1}{3} \times \frac{2}{5}}{\frac{1}{3} \times \frac{2}{15} + \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{1}{5}} = \frac{6}{11}.$$

A real number $\boldsymbol{X}$ connected with the outcome of a random experiment $\boldsymbol{E}$.

For example, if $\boldsymbol{E}$ consists of two tosses, we assign a number, counting number of **heads** (0,1 or 2).

Outcome	:	HH	HT	TH	TT
Value of $\boldsymbol{X}$	:	2	1	1	0

Thus to each outcome $\boldsymbol{\omega}$, there corresponds a real number $X(\boldsymbol{\omega})$.

Random Variable

Definition: A random variable is a function that associates a real number with each element in the sample space.

Example: Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable: Y, where Y is the number of red balls, are

Sample Space	y
RR	2
RB	1
BR	1
BB	0

Note: One-dimensional *r.v.* will be denoted by capital letters, X, Y, Z….etc.

The values which *X, Y, Z….etc*, can assume are denoted by lower case letters., *x, y, z…etc.*

Discrete Random Variable

Let ***X*** be a discrete random variable on a sample space ***S***, that is, ***X*** assigns only finite number or countably infinite number of values to ***S***.

Say, $R_X = \{x_1, x_2, \ldots\ldots, x_n, \ldots\ldots, \infty\}$

Example*:* marks obtained in a test, number of accidents per month, number of telephone calls per unit time, number of successes in n trials and so on.

Then, ***X*** induces a function ***p*** which assigns probabilities to the points in $\boldsymbol{R_X}$ as follows:

$$p(x_i) = p_X(x_i) = P(X = x_i) = P\{s \in S: X(s) = x_i\} \text{ for } i = 1, 2, \ldots., n$$

Probability Mass function: If X is a discrete random variable with distinct values $x_1, x_2, \ldots \ldots, x_n$ then the function $\boldsymbol{p_X(x)}$ is defined as : $\boldsymbol{p_X}(x_i)=P(X = x_i) = p_i$.

p is called the ***probability mass function*** of random variable ***X***.

The numbers $p(x_i); i = 1, 2, \dots\dots$ must satisfy the following conditions:

(i) $p(x_i) \geq 0 \ \forall \ i$

(ii) $\sum_{i=1}^{n} p(x_i) = 1$

Probability distribution :

The collection of pairs $\{x_i, p_i\}$ is called the probability distribution of the R.V. X.

$X = x_i$	x_1	....	x_i
p_i	p_1	....	p_i

Example:

I have an unfair coin for which $P(H) = p$, where $0 < p < 1$. I toss the coin repeatedly until I observe a heads for the first time. Let Y be the total number of coin tosses. Find the distribution of Y.

Answer:

First, we note that the random variable Y can potentially take any positive integer, so we have $R_Y = \mathbb{N} = \{1, 2, 3, \dots\}$. To find the distribution of Y, we need to find $P_Y(k) = P(Y = k)$ for $k = 1, 2, 3, \dots$. We have

$$P_Y(1) = P(Y = 1) = P(H) = p,$$

$$P_Y(2) = P(Y = 2) = P(TH) = (1-p)p,$$

$$P_Y(3) = P(Y = 3) = P(TTH) = (1-p)^2 p,$$

$$\begin{array}{cccc} . & . & . & . \\ . & . & . & . \\ . & . & . & . \end{array}$$

$$P_Y(k) = P(Y = k) = P(TT\dots TH) = (1-p)^{k-1} p.$$

Thus, we can write the PMF of Y in the following way

$$P_Y(y) = \begin{cases} (1-p)^{y-1} p & \text{for } y = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

2. If $p = \frac{1}{2}$, find $P(2 \leq Y < 5)$.

Answer:

2. if $p = \frac{1}{2}$, to find P$(2 \leq Y < 5)$, we can write

$$\begin{aligned} \mathrm{P}(2 \leq Y < 5) &= \textstyle\sum_{k=2}^{4} P_Y(k) \\ &= \textstyle\sum_{k=2}^{4} (1-p)^{k-1} p \\ &= \frac{1}{2}\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) \\ &= \frac{7}{16}. \end{aligned}$$

Continuous Random Variables

A random variable X is said to be a continuous random variable if it takes all possible values between certain limits or in an interval which may be finite or infinite.

Example: operating time between two failures of a computer

Probability density function (p.d.f):

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X, defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in R$,
2. $\int_{-\infty}^{\infty} f(x)\, dx = 1$,
3. $P(a < X < b) = \int_a^b f(x)\, dx$.

Distribution Function (or) Cumulative Distribution Function (cdf)

If X is a R.V, discrete or continuous, then $F(x) = P(X \leq x)$ is called the cumulative distribution function of X.

i) If X is discrete then $F(x) = \sum_{x_j \leq x} p_j$

ii) If X is continuous then $F(x) = P(-\infty \leq X \leq x) = \int_{-\infty}^{x} f(x)dx$

Relationship between cumulative distribution function and probability distribution function

$$f(x) = \frac{dF(x)}{dx}$$

If *p(x)* is the p.m.f of a random variable '*X*' then

$$Mean = \sum_{n=0}^{\infty} xp(x)$$

Variance $=\sum_{n=0}^{\infty}(x - mean)^2 p(x)$

Moment about the mean $= \sum_{n=0}^{\infty}(x - mean)^r p(x)$

If *f (x)* is the p.d.f of a random variable 'X' which defined in the interval (a, b) then

(i) Mean $= \int_a^b xf(x)dx$ (ii) Variance $= \int_a^b (x - mean)^2 f(x)dx$

(iii) Moment about mean $= \int_a^b (x - mean)^r f(x)dx$

Problem 1:

If the random variable takes the values 1,2,3 and 4 such that $2P(X=1)=3P(X=2)=P(X=3)=5P(X=4)$, find the probability distribution and the cumulative distribution function of X.

Solution:

Let $2P(X=1)=3P(X=2)=P(X=3)=5P(X=4)=k$

That is, $P(X=1)=\frac{k}{2}$, $P(X=2)=\frac{k}{3}$, $P(X=3)=k$, $P(X=4)=\frac{k}{5}$

We know that, $\sum_i p_i = 1$

That is, $\frac{k}{2}+\frac{k}{3}+k+\frac{k}{5}=1 \Rightarrow k=\frac{30}{61}$

The probability distribution of X is given by

x_i	1	2	3	4
$P(x_i)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

The c.d.f. F(x) is defined as $F(x) = P(X \le x)$.

When $x = 1, F(1) = P(X = 1) = \frac{15}{61}$

When x $= 2, F(2) = P(X = 1) + P(X = 2) = \frac{25}{61}$

When x $= 3, F(3) = P(X = 1) + P(X = 2) + P(X = 3) = \frac{55}{61}$

When $x = 4, F(4) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$

Problem 2:

A random variable X has the following probability function.

Values of X	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
$P(X=x)$	**a**	**3a**	**5a**	**7a**	**9a**	**11a**	**13a**	**15a**	**17a**

(i) **Find the value of a** (ii) **Find** $P(X<3)$, $P(0<X<3)$, $P(X\geq 3)$

(iii) Find cumulative distribution function of X.

Solution:

i) We know that , $\sum_i p_i = 1$

a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1

$81a = 1 \quad \Rightarrow a = \frac{1}{81}$

ii) $P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = a + 3a + 5a$

$= 9a = \frac{1}{9}$

$P(0 < X < 3) = P(X = 1) + P(X = 2) = 3a + 5a = 8a = \frac{8}{81}$

$P(X \geq 3) = 1 - P(X < 3) = 1 - \frac{1}{9} = \frac{8}{9}$

iii) Distribution function F(x) of X

Values of X	0	1	2	3	4	5	6	7	8
$F(x) = P(X \leq x)$	a $\frac{1}{81}$	4a $\frac{4}{81}$	9a $\frac{9}{81}$	16a $\frac{16}{81}$	25a $\frac{25}{81}$	36a $\frac{36}{81}$	49a $\frac{49}{81}$	64a $\frac{64}{81}$	81a 1

Problem 3: Let X be a random variable with PDF given by

$$f_X(x) = \begin{cases} cx^2 & |x| \le 1 \\ 0 & \text{otherwise} \end{cases}$$

a. Find the constant c.
b. Find $E(X)$ and $\mathrm{Var}(X)$.
c. Find $P(X \ge \frac{1}{2})$.

The expection of X is given by

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$ where $f(x)$ is probability density function.

a. To find c, we can use $\int_{-\infty}^{\infty} f_X(u)du = 1$:

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f_X(u)du \\ &= \int_{-1}^{1} cu^2 du \\ &= \frac{2}{3}c. \end{aligned}$$

Thus, we must have $c = \frac{3}{2}$.

b. To find EX, we can write

$$\begin{aligned} EX &= \int_{-1}^{1} u f_X(u)du \\ &= \frac{3}{2}\int_{-1}^{1} u^3 du \\ &= 0. \end{aligned}$$

In fact, we could have guessed $EX = 0$ because the PDF is symmetric around $x = 0$. To find Var(X), we have

$$\begin{aligned} \mathrm{Var}(X) &= EX^2 - (EX)^2 = EX^2 \\ &= \int_{-1}^{1} u^2 f_X(u)du \\ &= \frac{3}{2}\int_{-1}^{1} u^4 du \\ &= \frac{3}{5}. \end{aligned}$$

c. To find $P(X \geq \frac{1}{2})$, we can write

$$P(X \geq \frac{1}{2}) = \frac{3}{2}\int_{\frac{1}{2}}^{1} x^2 dx = \frac{7}{16}.$$

Problem 4: Let X be a continuous random variable with PDF given by

$$f_X(x) = \frac{1}{2}e^{-|x|}, \qquad \text{for all } x \in \mathbb{R}.$$

If $Y = X^2$, find the CDF of Y.

Answer:

First, we note that $R_Y = [0, \infty)$. For $y \in [0, \infty)$, we have

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X^2 \leq y) \\ &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} \tfrac{1}{2}e^{-|x|}\,dx \\ &= \int_0^{\sqrt{y}} e^{-x}\,dx \\ &= 1 - e^{-\sqrt{y}}. \end{aligned}$$

Thus,

$$F_Y(y) = \begin{cases} 1 - e^{-\sqrt{y}} & y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Problem 5: Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 4x^3 & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(X \leq \frac{2}{3} | X > \frac{1}{3})$.

Answer:

We have

$$\begin{aligned} P(X \leq \tfrac{2}{3} | X > \tfrac{1}{3}) &= \frac{P(\frac{1}{3} < X \leq \frac{2}{3})}{P(X > \frac{1}{3})} \\ &= \frac{\int_{\frac{1}{3}}^{\frac{2}{3}} 4x^3 dx}{\int_{\frac{1}{3}}^{1} 4x^3 dx} \\ &= \frac{3}{16}. \end{aligned}$$

Problem 6: **EXAMPLE:** Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} C(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) What is the value of C?

(b) Find $P\{X > 1\}$.

SOLUTION **(a)** Since f is a probability density function, we must have that $\int_{-\infty}^{\infty} f(x)\,dx = 1$, implying that

$$C\int_0^2 (4x - 2x^2)\,dx = 1$$

or

$$C\left[2x^2 - \frac{2x^3}{3}\right]\Big|_{x=0}^{x=2} = 1$$

or

$$C = \tfrac{3}{8}$$

(b) Hence

$$P\{X > 1\} = \int_1^{\infty} f(x)\,dx = \tfrac{3}{8}\int_1^2 (4x - 2x^2)\,dx = \tfrac{1}{2} \quad ■$$

Problem 7:

If a random variable 'X' has the p.d.f. $f(x) = \begin{cases} \dfrac{(x+1)}{2} & ,-1 < x < 1 \\ 0 & ,otherwise \end{cases}$ Then find the mean and variance.

Answer:

We know that, Mean = $\int_{-1}^{1} xf(x)dx$ = $\int_{-1}^{1} x\dfrac{(x+1)}{2}dx$ = $\dfrac{1}{2}\int_{-1}^{1}(x^2+x)dx$ = $\dfrac{1}{3}$

Variance = $\int_{-1}^{1}(x-mean)^2 f(x)dx$ = $\int_{-1}^{1}(x-\dfrac{1}{3})^2\dfrac{(x+1)}{2}dx$ $= \int_{-1}^{1}(x^2-\frac{2}{3}x+\frac{1}{9})\frac{(x+1)}{2}dx$

$= \frac{1}{2}\int_{-1}^{1}(x^3-\frac{2}{3}x^2+9x+x^2-\frac{2}{3}x+\frac{1}{9}) = \frac{1}{2}\int_{-1}^{1}(x^3+\frac{1}{3}x^2+\frac{25}{3}x+\frac{1}{9})dx = \frac{1}{2}\left[\frac{x^4}{4}+\frac{x^3}{9}+\frac{25}{3}x^2+\frac{x}{9}\right]_{-1}^{1} = \frac{1}{2}\left[2\cdot\frac{1}{9}+2\cdot\frac{1}{9}\right] = \frac{2}{9}.$

Two-dimensional Random Variable

Our study of random variables and their probability distributions in the preceding sections is restricted to one-dimensional sample spaces.
In that we recorded outcomes of an experiment as values assumed by a single random variable.
There will be situations, however, where we may find it desirable to record the simultaneous outcomes of several random variables

Example:

For instance, blood pressure and cholesterol for each individual are measured simultaneously.

In a study to determine the likelihood of success in college based on high school data, we might use a three dimensional sample space and record for each individual his or her aptitude test score, high school class rank, and grade-point average at the end of freshman year in college.

Definition: Let S be a sample space associated with a random experiment E. Let X and Y be two random variables defined on S. then the pair (X,Y) is called a Two–dimensional random variable.
The value of (X,Y) at a point $s \in S$ is given by the ordered pair of real numbers (X(s),Y(s))=(x,y) where X(s)=x, Y(s)=y.

Two –Dimensional discrete random variable: If the possible values of (X,Y) are finite or countably infinite, then (X,Y) is called a two-dimensional discrete random variable.
When (X,Y) is a two-dimensional discrete random variable the possible values of (X,Y) may be represented as (x_i, y_j), $i = 1, 2, 3, \dots n,\ \ j = 1, 2, 3, \dots m$.

Example: Consider the experiment of tossing a coin twice.

The sample space is S = {HH, HT, TH, TT}.Let X denotes the number of heads obtained in the first toss and Y denote the number of heads in the second toss.

s	HH	HT	TH	TT
X(s)	1	1	0	0
Y(s)	1	0	1	0

(X, Y) is a two-dimensional random variable or bi-variate random variable. The range space of (X, Y) is {(1,1), (1,0), (0,1), (0,0)} which is finite and so (X, Y) is a two-dimensional discrete random variables.

Joint probability distribution of Discrete R.V

The function $f(x, y)$ is a **joint probability distribution** or **probability mass function** of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y),
2. $\sum_x \sum_y f(x, y) = 1$,
3. $P(X = x, Y = y) = f(x, y)$.

$X \backslash Y$	y_1	y_2	y_3	$\cdots$	y_j	$\cdots$	y_m	Total
x_1	p_{11}	p_{12}	p_{13}	$\cdots$	p_{1j}	$\cdots$	p_{1m}	$p_{1\cdot}$
x_2	p_{21}	p_{22}	p_{23}	$\cdots$	p_{2j}	$\cdots$	p_{2m}	$p_{2\cdot}$
x_3	p_{31}	p_{32}	p_{33}	$\cdots$	p_{3j}	$\cdots$	p_{3m}	$p_{3\cdot}$
.	.	.	.	$\cdots$	.	$\cdots$	.	
.	.	.	.	$\cdots$	.	$\cdots$	.	$\vdots$
.	.	.	.	$\cdots$	.	$\cdots$	.	
x_i	p_{i1}	p_{i2}	p_{i3}	$\cdots$	p_{ij}	$\cdots$	p_{im}	$p_{i\cdot}$
.	.	.	.	$\cdots$	.	$\cdots$	.	
.	.	.	.	$\cdots$	.	$\cdots$	.	$\vdots$
.	.	.	.	$\cdots$	.	$\cdots$	.	
x_n	p_{n1}	p_{n2}	p_{n3}	$\cdots$	p_{nj}	$\cdots$	p_{nm}	$p_{n\cdot}$
Total	$p_{\cdot 1}$	$p_{\cdot 2}$	$p_{\cdot 3}$	$\cdots$	$p_{\cdot j}$	$\cdots$	$p_{\cdot m}$	1

Roll two dice. Let X be the value on the first die and let T be the total on both dice. Here is the joint probability table:

$X \backslash T$	2	3	4	5	6	7	8	9	10	11	12
1	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0	0
2	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0	0
3	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0	0
4	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0	0
5	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36	0
6	0	0	0	0	0	1/36	1/36	1/36	1/36	1/36	1/36

Exercise:

Suppose a car showroom has 10 cars of particular brand out of which 5 are good, 2 have defective transmission 3 have defective steering. If 2 cars are selected at random, find the joint probability distribution table. Let X denotes the number of cars with DT Y denotes the number of cars with DS, find the joint probability table.

Answer:

$$P(X=0, Y=0) = \frac{\binom{5}{2}}{\binom{10}{2}} = \frac{2}{9}$$

2 choices

1 good transmission 1 DS

$$P(X=0, Y=1) = \frac{\binom{5}{1}\binom{3}{1}}{\binom{10}{2}} = \frac{1}{3}$$

and so on….

$$P(X=0, Y=2) = \frac{{}^{5}C_{0}\ {}^{3}C_{2}}{{}^{10}C_{2}} = \frac{1 \times 3}{\frac{10 \times 9}{2}} = \frac{1}{15}$$

no choice 2 DS left for DT

X \ Y	0	1	2
0	2/9	1/3	1/15
1	2/9	2/15	0
2	1/45	0	0

Joint probability distribution of Continuous R.V

The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y),
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y)\, dx\, dy$, for any region A in the xy plane.

Exercise:

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \le x \le 1, 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Verify condition 2

(b) Find $P[(X,Y) \in A]$, where $A = \{(x,y) \mid 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$.

Answer:

(a) The integration of $f(x,y)$ over the whole region is

$$\begin{aligned}
\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy &= \int_0^1\int_0^1 \frac{2}{5}(2x+3y)\,dx\,dy \\
&= \int_0^1 \left(\frac{2x^2}{5} + \frac{6xy}{5}\right)\Bigg|_{x=0}^{x=1} dy \\
&= \int_0^1 \left(\frac{2}{5} + \frac{6y}{5}\right) dy = \left(\frac{2y}{5} + \frac{3y^2}{5}\right)\Bigg|_0^1 = \frac{2}{5} + \frac{3}{5} = 1.
\end{aligned}$$

(b) To calculate the probability, we use

$$
\begin{aligned}
P[(X,Y) \in A] &= P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right) \\
&= \int_{1/4}^{1/2} \int_{0}^{1/2} \frac{2}{5}(2x + 3y)\; dx\; dy \\
&= \int_{1/4}^{1/2} \left(\frac{2x^2}{5} + \frac{6xy}{5}\right)\Bigg|_{x=0}^{x=1/2} dy = \int_{1/4}^{1/2} \left(\frac{1}{10} + \frac{3y}{5}\right) dy \\
&= \left(\frac{y}{10} + \frac{3y^2}{10}\right)\Bigg|_{1/4}^{1/2} \\
&= \frac{1}{10}\left[\left(\frac{1}{2} + \frac{3}{4}\right) - \left(\frac{1}{4} + \frac{3}{16}\right)\right] = \frac{13}{160}.
\end{aligned}
$$

Marginal Distributions

Given the joint probability distribution $f(x, y)$ of the discrete random variables X and Y, the probability distribution $g(x)$ of X alone is obtained by summing $f(x, y)$ over the values of Y. Similarly, the probability distribution $h(y)$ of Y alone is obtained by summing $f(x, y)$ over the values of X.

Definition:

The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

for the continuous case.

Exercise: Find $g(x)$ and $h(y)$ for the following distribution table:

	$f(x,y)$	x = 0	1	2	Row Totals $h(y)$
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$g(x)$: Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Answer:

For the random variable X, we see that

$$g(0) = f(0,0) + f(0,1) + f(0,2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1,0) + f(1,1) + f(1,2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2,0) + f(2,1) + f(2,2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table . In a similar manner we could show that the values of $h(y)$ are given by the row totals. In tabular form, these marginal distributions may be written as follows:

x	0	1	2
$g(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

y	0	1	2
$h(y)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$

Exercise:

Find $g(x)$ and $h(y)$ for the joint density function of

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \le x \le 1, 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Answer:

By definition,

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\, dy = \int_0^1 \frac{2}{5}(2x+3y)\, dy = \left(\frac{4xy}{5} + \frac{6y^2}{10}\right)\Bigg|_{y=0}^{y=1} = \frac{4x+3}{5},$$

for $0 \le x \le 1$, and $g(x) = 0$ elsewhere. Similarly,

$$h(y) = \int_{-\infty}^{\infty} f(x,y)\, dx = \int_0^1 \frac{2}{5}(2x+3y)\, dx = \frac{2(1+3y)}{5},$$

for $0 \le y \le 1$, and $h(y) = 0$ elsewhere.